Name

Mathematics Advanced Year 11

The Derivative as a Rate of Change

Book 5

Version: 261024

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Contents

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Learning Intention ..... **Error! Bookmark not defined.**

Syllabus Content

MAV-11-06 interprets the meaning of the derivative and determines the derivative of functions to solve problems

The derivative as a rate of change

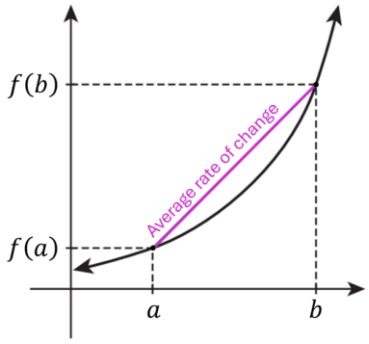
- Interpret $f'(c)$ as the instantaneous rate of change of the function $f(x)$ at $x = c$
- Define and distinguish between displacement and distance and between velocity and speed
- Use graphs of functions and their derivatives, without the use of algebraic techniques, to describe and interpret physical phenomena
- Use the notation $\frac{dx}{dt}$ or $\dot{x}$ to represent the velocity of a particle with displacement x from a point as a function of time t
- Solve problems by determining the velocity of a particle moving in a straight line, given its displacement from a point as a function of time

Average Rate of Change

ⓘ Average Rate of Change

The **average rate of change** between $x = a$ and $x = b$ is the **gradient of the secant**.

$$\begin{aligned}\text{Average rate of change} &= \frac{\Delta y}{\Delta x} \\ &= \frac{f(b) - f(a)}{b - a}\end{aligned}$$



Example Find average rate of change across an interval.

A container is being filled with water.
The volume V, in litres, after t minutes is given by

$$V(t) = t^2 + t + 1$$

Find the average rate of change of volume between $t = 1$ and $t = 5$

$$\begin{aligned}\frac{\Delta y}{\Delta x} &= \frac{V(5) - V(1)}{5 - 1} \\ &= \frac{31 - 3}{4} \\ &= 7 \text{ L/min}\end{aligned}$$

The temperature T, in °C, of a glass of water over t minutes is changing according to

$$T(t) = 30 \times 2^{-t} + 20$$

Find the average rate of change in temperature over the first two minutes.

$$\begin{aligned}\frac{\Delta y}{\Delta x} &= \frac{T(2) - T(0)}{2 - 0} \\ &= \frac{27.5 - 50}{2} \\ &= -11.25 \text{ °C/min}\end{aligned}$$

Practice

A model rocket is launched into the air. Its height after t seconds is given by $h(t) = -t^2 + 10t$, where h is in metres.

Find the average rate of change of height between $t = 0$ and $t = 4$.

6 m/s

Find the average rate of change of height between $t = 6$ and $t = 10$.

−6 m/s

1 $Q = 8t - t^2$

a Find the values of Q when $t = 1$ and $t = 3$.

7 and 15

b Hence find the average rate of change from $t = 1$ to $t = 3$.

4

c Similarly find the average rate of change from $t = 5$ to $t = 7$.

-4

2 A bucket full of water has a leak, and the volume after t minutes is given by

$$V(t) = t^2 - 4t + 4$$

where V is measured in litres.

a Find the average rate of change of V during the first minute.
Remember to have units in your answer.

-3 L/min

b Find the average rate of change of V over the first two minutes.

-2 L/min

c Explain why the model is only valid for t $\in$ [0, 2].
What happens to a leaky bucket eventually?

The bucket is empty at $t = 2$

3 The value of a car, in $, is modelled by the equation $V(t) = 50000(1.15)^{-t}$, where t is in years.
What is the average rate of change in the value of the car over the first five years?

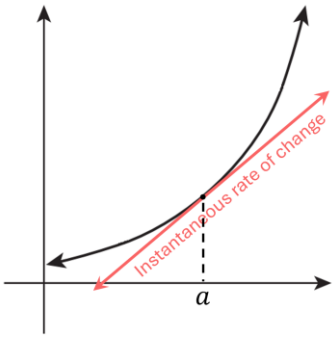
$-\$5028.23/\text{yr}$

Instantaneous Rate of Change

ⓘ Instantaneous Rate of Change

The **instantaneous rate of change** at $x = a$ is the **gradient of the tangent**.

$$\begin{aligned}\text{Instantaneous rate of change} &= \frac{dy}{dx} \\ &= f'(a)\end{aligned}$$



ⓘ **Identify** rate of change formula.

Find the formula for the rate of change of each function.

a $h = 50t^2 + 3t$

$\frac{dh}{dt} =$

b $A = (h + 2)^5$

c $V = 7(2t + 5)^3$

d $y = \sqrt{4 - x^2}$

e $S = 4\pi r^2$

f $P = \frac{5+2t}{10-t}$

ⓘ **Interpret** rate of change scenarios.

A ball shoots up and comes down. Its height is $h(t)$ where h is in metres and t in seconds. This is what each mathematical operation means.

Find $\boldsymbol{h(a)}$: Calculate the height at $t = a$ seconds.

Solve $\boldsymbol{h(t) = a}$: Find when the ball reaches a height of a metres.

Find $\boldsymbol{h'(t)}$: Find the rate of change of the height.
Find an equation describing how the height is increasing/decreasing.

Find $\boldsymbol{h'(a)}$: Find how quickly the ball's height changes at $t = a$ seconds.

Solve $\boldsymbol{h'(t) = a}$: Find when the height of the ball is changing/increasing/decreasing at a m/s.

Identify what the following questions are asking you to find.

a	What was the height of the ball when it was thrown?	**b**	At what rate was the height changing 20 seconds after being thrown?
c	When was the ball at maximum height?	**d**	When was the ball 20 m high?
e	When was the ball's height increasing by 20 m/s?	**f**	What is the formula that describes the rate of change over time?

The volume of water in a leaky tank is given by $V(t)$, where V is in litres and t in minutes.

a	How fast was the water leaking out after 5 minutes?	**b**	How much water was in the tank initially?
c	How long did it take for the tank to become completely empty?	**d**	When was the volume decreasing by 20 L/min?

The height of a bungee jumper is given by $h(t)$, where h is in metres and t in seconds.

a	What was the height of the bungee jumper after 20 seconds?	**b**	When was the bungee jumper's height increasing by 20 m/sec?
c	When was the bungee jumper's height minimum?	**d**	What was the minimum height of the bungee jumper?

Example Solve rate of change problems.

A ball is thrown into the air; its height is given by the function $h(t) = -5t^2 + 20t + 1$ where h is in metres and t in seconds.

a Find the initial height of the ball.

$h(0) = -5(0)^2 + 20(0) + 1$

$= 1$ m

b Find the initial rate at which the height was changing.

$h'(t) = -10t + 20$

$h'(0) = -10(0) + 20$

$= 20$ m/s

c Find the height after 3 seconds.

$h(3) = -5(3)^2 + 20(3) + 1$

$= 16$ m

d Find the rate at which the height is changing after 3 seconds.

$h'(3) = -10(3) + 20$

$= -10$ m/s

e Find when the ball was at the maximum height.

$h'(t) = 0$

$-10t + 20 = 0$

$\therefore t = 2$ s

f Find the maximum height of the ball.

$h(2) = -5(2)^2 + 20(2) + 1$

$= 21$ m

Practice

An infestation of ants hit a Year 7's locker but was gradually brought under control. The ant population P, in hundreds, t hours after it was discovered, is $P(t) = 7 + 6t - t^2$.

a Find the initial population of the ants.

$P(0) = 7$ hundred

b Find the initial rate at which the population was changing.

$P'(t) = 6 - 2t$

$P'(0) = 6$ hundred per hour

c Find the rate at which the ant population is changing after 4 hours.

$P'(4) = -2$ hundred per hour

d Find when the ant population was at its maximum.

$P'(t) = 0$

$t = 3$ hours

e Find the maximum population of the ants.

$P(3) = 16$ hundred

f When were the ants finally eliminated?

$P(t) = 0$

$t = -1$ or 7

They were eliminated after 7 hours.

1 Find the formula for the rate of change for each function.

a $h = 20t - 4t^2$

$\frac{dh}{dt} = 20 - 8t$

b $D = 5t^3 + 2t^2 + 1$

$\frac{dD}{dt} = 15t^2 + 4t$

c $A = 16x - 2x^2$

$\frac{dA}{dx} = 16 - 4x$

d $x = 3t^5 - t^4 + 2t - 3$

$\frac{dx}{dt} = 15t^4 - 4t^3 + 2$

e $V = \frac{4}{3}\pi r^3$

$\frac{dV}{dr} = 4\pi r^2$

f $S = 2\pi r + \frac{50}{r^2}$

$\frac{dS}{dr} = 2\pi - \frac{100}{r^3}$

g $D = \sqrt{x^2 - 4}$

$\frac{dD}{dx} = \frac{x}{\sqrt{x^2 - 4}}$

h $S = 800r + \frac{400}{r}$

$\frac{dS}{dr} = 800 - \frac{400}{r^2}$

2 If $h = t^3 - 7t + 5$, find:

a the average rate of change of h between $t = 3$ and $t = 4$.

30

b the instantaneous rate of change of h when $t = 3$.

20

3 The volume of water V in litres flowing through a pipe after t seconds is given by $V = t^2 + 3t$. Find the rate at which the water is flowing when $t = 5$.

13 L/s

4 The mass in grams of a melting ice block is given by the formula $M = t - 2t^2 + 100$, where t is time in minutes.

a Find the average rate of change at which the ice block is melting between:

i 1 and 3 minutes

–7 g/min

ii 2 and 5 minutes.

–13 g/min

b Find the rate at which it is melting at 5 minutes.

–19 g/min

5 The surface area in cm² of a balloon being inflated is given by $S = t^3 - 2t^2 + 5t + 2$, where t is time in seconds. Find the rate of increase in the balloon's surface area at 8 s.

165 cm² /s

6 A circular disc expands as it is heated. The area, in cm², of the disc increases according to the formula $A = 4t^2 + t$, where t is time in minutes. Find the rate of increase in the area after 5 minutes.

41 cm²/min

7 Orange juice is being poured at a constant rate into a glass. After t seconds there are V mL of juice in the glass, where $V = 60t$.

a How much juice is in the glass after 3 seconds?

180 mL

b Show that the glass was empty to begin with.

$V(0) = 60(0) = 0$

c If the glass takes 5 seconds to fill, what is its capacity?

$V(5) = 300$ mL

d Differentiate to find the rate at which the glass is being filled.

$\frac{dV}{dt} = 60$ mL/s

e What sort of function is the derivative in part **d**?

The derivative is a constant function.

8 The volume V litres of fuel in a tanker, t minutes after it has started to empty, is given by $V = 200(400 - t^2)$. Initially the tanker was full.

a Find the initial volume of fuel in the tanker.

80 000 L

b Find the volume of fuel in the tanker after 15 minutes.

35 000 L

c Find the time taken for the tanker to empty.

20 min

d Show that $\frac{dV}{dt} = -400t$, and hence find the rate at which the tanker is emptying after 5 minutes.

Decreasing by 2000 L/min

9 The volume of water in a container can be modelled by the equation

$$V = \frac{10 - t}{2 + t}$$

where t is time in minutes and V is measured in litres.

a How much water was in the container initially?

5 L

b Show that $V'(t) = -\frac{12}{(2+t)^2}$

Using quotient rule

$V'(t) = \frac{-1(2+t) - (10-t)(1)}{(2+t)^2}$

$= -\frac{12}{(2+t)^2}$

c Explain why the container is always losing water.

The derivative $-\frac{12}{(2+t)^2}$ is always negative

d Find the rate at which V is decreasing initially.

3 L/min

e Find the rate at which V is decreasing when $t = 2$.

0.75 L/min

10 A ball is dropped from the top of a building, and the distance of the ball from the ground after t seconds is given by the equation $h = 100 - 5t^2$, where h is measured in metres.

a How tall is the building?

100 m

b How long does it take for the ball to hit the ground?

$\sqrt{20}$ = 4.47 s

c Hence, state the time interval $t \in [a, b]$ where this model is valid.

$t \in [0, \sqrt{20}]$

d What is the rate of change $\frac{dh}{dt}$ of the height:

i initially?

0 m/s

ii when it hits the ground?

$-10\sqrt{20}$ m/s

e When is the rate at which h decreases the largest?

$t = 4.47$, just when it is about to hit the ground.

11 A circular disc expands as it is being heated. The area of the disc after t minutes is given by $A = 3t^2 + t$, where A is in cm².

a Find the rate at which the area increases after 4 minutes.

25 cm² /min

b Find the average rate of change during the 4th minute. (4th minute is between $t = 3$ and 4).

22 cm² /min

12 The rate of water flow into a tank is given by $\frac{dV}{dt} = 10(2 - t)$, where $\frac{dV}{dt}$ is measured in litres per minute. The tank initially had 60 litres of water in it.

a Describe the difference in the water flow when $t = 1$ and $t = 4$.
In your answer, refer to the direction of water flow.

The rate is positive when $t = 1$ (water is flowing in)

The rate is negative when $t = 4$ (water is flowing out)

b Show that the volume of water increases for a certain amount of time, and then decreases afterwards.

Consider the derivative $10(2 - t)$, it is a decreasing straight line that is positive when $t < 2$ and negative when $t > 2$.

When $t < 2$, volume is increasing, when $t > 2$, the volume is decreasing.

c Show that the formula $V = 20t - 5t^2 + 60$ satisfies the given information.

$\frac{dV}{dt} = 20 - 10t = 10(2 - t)$

13 A water tank is being emptied. The quantity Q litres of water remaining in the tank at any time t minutes after it starts to empty is given by $Q(t) = 1000(20 - t)^2, 0 \leq t \leq 20$.

a At what rate is the tank being emptied at any time t?

$2000(20 - t)$ L/min

b How much time does it take to empty the tank? (When is $Q = 0$?)

After 20 minutes

c At what time is the water flowing out at a rate of 20 000 litres per minute?

After 10 minutes

d What is the average rate at which the water flows out in the first 5 minutes?

35 000 L/min

14 A block of ice of mass 8 kg melts according to the formula $M = 8 - \frac{t^2}{32}$, where M is the mass remaining, in kilograms, after t minutes.

a Find the time taken for all the ice to melt.

16 minutes

b Find the rate at which the ice is melting after 4 minutes.

–0.25 kg/min

c Find the rate at which the ice is melting when 2 kg of ice has melted.

–0.5 kg/min

d Is the melting rate faster initially, or just before it is gone completely?

It is faster just before it's gone.

15 Water is poured into a tank and the volume V, in litres, after t seconds is $V = 200 + 80t - 2t^2$.

a Find the amount of water initially.

200 L

b If the tank was full in 20 seconds, what is the capacity of the tank?

1000 L

c Find the average rate of change during the first 10 seconds and compare it with the instantaneous rate of change after 10 seconds.

60 L/s vs 40 L/s

d When was the flow rate the greatest?

At $t = 0$

16 Water surges into a rock pool, and then out again. The mass M in kilograms of water in the pool t seconds after time zero is given by the function $M = 10t - t^2$.

a Find the rate $\frac{dM}{dt}$ as a function of time t.

$\frac{dM}{dt} = 10 - 2t$

b Find (with units) the values of M and $\frac{dM}{dt}$ when $t = 4$.

$M = 24$ kg, $\frac{dM}{dt} = 2$ kg/s

c Find the value of M when $t = 2$, and hence find the average rate of change over the time interval from $t = 2$ to $t = 4$.

$M = 16$ kg, avg rate of change $= 4$ kg/s

d At what times is M zero?

$M = t(10 - t)$,
0 and 10 seconds

e Therefore, for how long was there water in the pool?

10 seconds

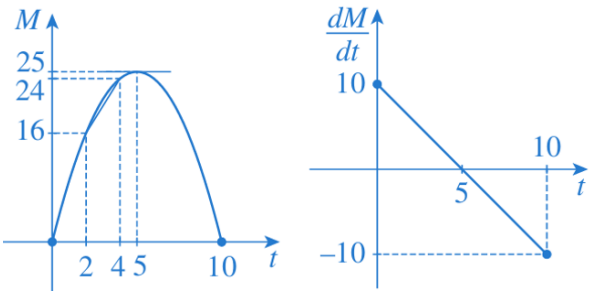
f How long after time zero is $\frac{dM}{dt}$ zero?

$\frac{dM}{dt} = 10 - 2t = 0$
$t = 5$ seconds

g For how long was the water level in the pool rising, and for how long was it falling?

$\frac{dM}{dt} > 0$ for the first 5 seconds and < 0 for the next 5 seconds

h Sketch the graphs of M and $\frac{dM}{dt}$ as functions of t, showing these results.
Add the chord corresponding to the result in part **c** and the tangent corresponding to the result in part **f**.



17 The share price $\$P$ of a bank t years after it opened on 1 Jan 1970 was $P = -0.4t^2 + 4t + 2$.

a What was the initial share price?

$\$2$

b What was the share price after one year?

$\$5.60$

c At what rate was the share price increasing after two years?

$\frac{dP}{dt} = -0.8t + 4$

$\$2.40/\text{yr}$

d Find when the share price was stationary.

$t = 5$, at the start of 1975

e Explain why the maximum share price was \$12, at the start of 1975.

$t = 5$ is the turning point of P.

The share price was increasing before then, and decreasing after.

f Explain why the rate of change of the price decreased at a constant rate.

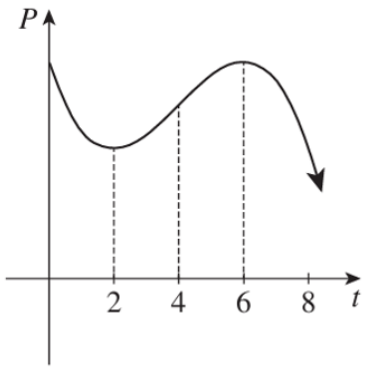
$\frac{dP}{dt}$ is linear with negative gradient -0.8

g The directors decided to close the bank when the share price fell back to its initial value. When did this happen?

By symmetry of a parabola, it would take another 5 years to go from the maximum back to the starting value.

∴ At the start of 1980.

18 The graph shows the share price P in a company t months after the start of the year.



a When is the price stationary, neither increasing/decreasing?

$t = 2, 6$

b When is the price maximum?

$t = 0, 6$

c When is the price increasing and when is it decreasing?

Increasing $2 < t < 6$, decreasing $0 < t < 2$ and $t > 6$

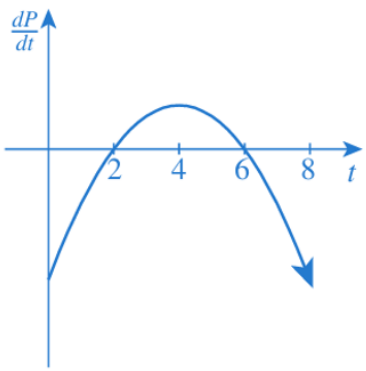
d When is the share price increasing most rapidly?

$t = 4$

e When is the share price increasing at an increasing rate?

Between $t = 2$ and $t = 4$

f Sketch the graph of $\frac{dP}{dt}$ as a function of time t.



19 A lightning bolt hits the ground with a tremendous noise. The noise spreads out across the suburbs in a circle of area A, startling people.

a The speed of sound in air is about $\frac{1}{3}$ km/s. Show that the area A in which people are startled as a function of the time t in seconds after the strike is $A = \frac{\pi}{9}t^2$.

Using distance = speed × time:

$$r = \frac{1}{3}t$$

$$A = \pi r^2 = \pi\left(\frac{t}{3}\right)^2 = \frac{\pi}{9}t^2$$

b Find the rate at which the area of startled people is increasing.

$$\frac{dA}{dt} = \frac{2\pi}{9}t$$

c Find, correct to four significant figures, the time when the area is 5 km².

$$t = \sqrt{\frac{45}{\pi}} \approx 3.785 \text{ s}$$

d Find the rate at which the area is increasing at that time

$$\frac{dA}{dt} \text{ when } t = \sqrt{\frac{45}{\pi}}$$

$$= 2.642 \text{ km}^2/\text{s}$$

20 HSC Advanced Sample Question Band 4

A magpie plague hit Sydney but was eventually brought under control. A researcher estimated that the population M, in hundreds, t months after 1st January, was $M = 7 + 20t - 3t^2$.

a Find the magpie population on 1st March.

$M(2) = 35$, 3500 magpies

b At what rate was the population changing at this time?

$$M'(t) = 20 - 6t$$

$$M'(2) = 8$$

Increasing by 800 magpies per month

c In what month does the magpie population start to decrease?

Find turning point

$M'(t) = 0$ when $t = \frac{10}{3} = 3\frac{1}{3}$

Population starts to decrease in April, since $t = 0$ is January

Applications of Differentiation to Motion

ⓘ Displacement, Velocity, Acceleration

- **Magnitude** is the distance from zero.
- **Direction** is positive or negative.
- **Distance** and **speed** are called **scalar** quantities, they only have magnitude, no direction.
- **Displacement**, **velocity**, and **acceleration** are **vector** quantities.
  They have magnitude *and* direction.
- In questions, it is important to define the positive direction to be able to interpret results.
  e.g. up is positive, down is negative; or right is positive, left is negative.

Displacement	Velocity	Acceleration
Position relative to a starting point.	Rate of change of displacement over time.	Rate of change of velocity over time
x	v or $\dot{x}$ or $\frac{dx}{dt}$	a or $\ddot{x}$ or $\frac{dv}{dt}$
E.g. 3 metres north.	E.g. 5 m/s west.	E.g. 9.8 m/s^2 down.
Distance is the magnitude.	Speed is the magnitude.	

ⓘ Language of Motion

Initial:	$t = 0$	*At rest/stationary:*	$v = 0$
At the origin:	$x = 0$	*Constant velocity:*	$a = 0$

ⓘ **Interpret** language of motion

Identify what the question is asking you to find, answer in function notation in terms of $x, \dot{x},$ or $\ddot{x}$.

The movement of a particle over time is given by $x(t)$.

a Find the initial displacement.	**b** Find when the particle is stationary.
c Find the initial velocity.	**d** Find when the particle is at the origin.
e Find when the particle is at rest.	**f** Find the initial acceleration.

Example Solve motion problems involving differentiation

A ball is dropped from a platform, its vertical displacement is $x = 50 - 5t^2$, where x is in metres and t is in seconds.

a Find the initial displacement. What does this represent?

$x(0) = 50$ m.

This is the height of the platform.

b Find its displacement at 2 s.

$x(2) = 50 - 5(2)^2$

$= 30$ m.

c Find its velocity at 2 s.

$\dot{x} = -10t$

$\dot{x}(2) = -20$ m/s

d Find its acceleration at 2 s. What do you notice about the acceleration?

$\ddot{x} = -10$

$\ddot{x}(2) = -10$ m/s^2

The acceleration doesn't change.

Practice

A rocket is launched upwards. Its vertical displacement is $x = -5t^2 + 20t + 5$.

a Find the initial displacement.

5 m

b Find the velocity after 3 seconds.

-10 m/s

c Find when the rocket is stationary. What is the significance of this point?

$t = 2$

This is the maximum height of the rocket.

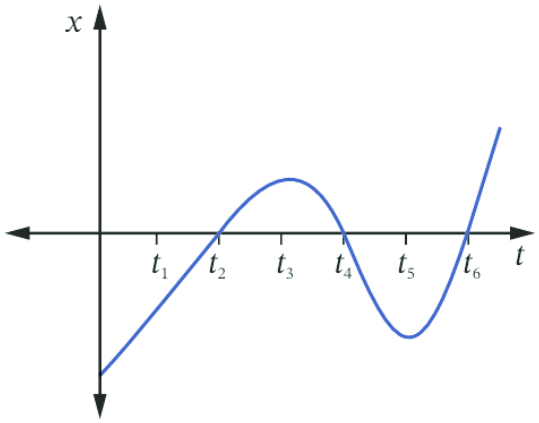
d Find the acceleration of the rocket.

-10 m/s^2

Example Interpret graphs of motion

This graph shows the displacement of a particle from the origin as it moves in a straight line.

a When is the particle at rest?
b When is it at the origin?
c When is it moving at its greatest speed?



a) The particle is at rest when $v = 0$.

The gradient is zero at t_3 and t_5

b) The particle is at the origin when $x = 0$

$x = 0$ at t_2, t_4, t_6

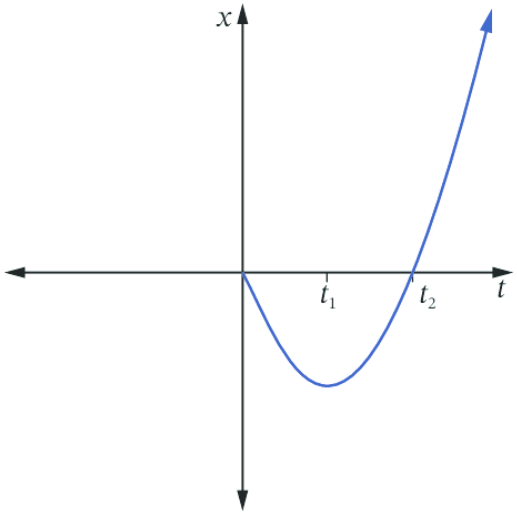
c) The speed is greatest when the gradient is steepest.

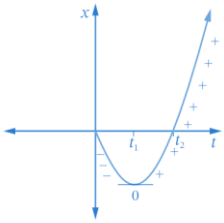
By inspecting the curve, this happens at around t_4.

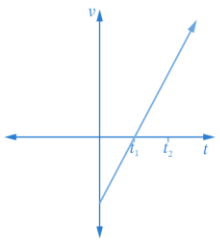
Practice

This graph shows the displacement of a particle from the origin as it moves in a straight line.

a Find when the particle is at rest.
b Find when the particle is at the origin.
c What happens to the velocity as $t \to \infty$?
d Sketch a graph of the velocity.







t_1
$0, t_2$
$v \to \infty$ as $t \to \infty$

1 A car is d km from home after t hours according to the formula $d = 10t^2 + 5t + 11$.

a How far is the car from home:

i initially?

11 km

ii after 3 hours?

116 km

b At what speed is the car travelling after:

i 3 hours?

65 km/h

ii 5 hours?

105 km/h

2 The displacement of a particle is $x = t^3 - 9t$ cm, where t is time in seconds.

a Find the velocity of the particle at 3 s.

18 cm/s

b Find the acceleration at 2 s.

12 cm/s²

c Show that the particle is initially at the origin, and find any other times that the particle will be at the origin.

When $t = 0, x = 0$

At $t = 3$

d At what time will the acceleration be 30 cm s^{-2}?

$t = 5$

3 A particle is moving with displacement $x = 2t^2 - 8t + 3$, where x is in metres and t is in seconds.

a Find its initial velocity.

$\dot{x}(0) = -8$ m/s

b Show that its acceleration is constant and find its value.

$\ddot{x} = 4$, constant acceleration of 4 m/s^2

c Find its displacement at 5 s.

13 m

d Find when the particle's velocity is zero.

2 s

e What will the particle's displacement be at that time?

–5 m

4 A stone is dropped from the top of a tall building. The stone's height h metres above the ground t seconds later is $h = 80 - 5t^2$.

a How high is the building?

$h(0) = 80$. 80 metres

b How many seconds does the stone take to reach the ground?

$h = 0$ when $t = 4$, so 4 seconds

c The velocity v is the rate of change $\frac{dh}{dt}$ of the height. Find the velocity function.

$v = \frac{dh}{dt} = -10t$

d How fast is the stone travelling when it hits the ground?

At $t = 4$, $v(4) = -40$. It hits the ground at 40 m/s

e Acceleration is the rate of change dv/dt of the velocity. Find the acceleration of the stone (the units are 'metres-per-second, per second', usually written as m/s²).

$\frac{dv}{dt} = -10$ m/s²

5 A particle moves in a straight line so that its displacement x m from a fixed point O on the line at any time t seconds ($t \geq 0$) is given by $x = t^2 - 5t + 6$. Find:

a its initial displacement.

6 m

b its initial velocity.

–5 m/s

c when it first passes through O and with what velocity.

$t = 2$, $v = -1$ m/s

d when it passes through O the second time and with what velocity.

$t = 3, v = 1$ m/s

e when and where its velocity is zero.

$t = 2.5$, $x = -0.25$ m

6 The displacement x m at time t seconds ($t \geq 0$) of a particle moving in a straight line is given by $x = t^2 - 5t + 4$.

a At what time is its velocity zero?

$t = 2.5$ s

b What is the acceleration at this time?

$a = 2$ m/s²

c What is the distance travelled in the first 4 seconds?

8.5 m

d At what time is the velocity 8 m s⁻¹?

6.5 s

7 A point moving in a straight line is distant x m from the origin O at time t, where $x = 2t^3 - 15t^2 + 36t$.

a Find the velocity and acceleration at any time t.

$\dot{x} = 6t^2 - 30t + 36$

$\ddot{x} = 12t - 30$

b Find the initial velocity and acceleration.

36 m/s, –30 m/s²

c At what times is the velocity zero?

$t = 2, 3$ s

d At what time is the acceleration zero? Find the velocity and position at this time.

$t = 2.5$ s

$\dot{x}(2.5) = -1.5$ m/s

$x(2.5) = 27.5$ m

e During what interval of time is the velocity negative?

$6t^2 - 30t + 36 < 0$

$(t - 3)(t - 2) < 0$

$2 < t < 3$

8 Two bodies move along a straight path, starting at the same time, so their displacement x m from a fixed point O at any time t is given by $x = t + 6$ and $x = t^2 + 4$ respectively.
At what times are they:

a Together.

$t + 6 = t^2 + 4$

$t = 2$

b Travelling with the same velocity?

Velocities of each particle are $\dot{x} = 1$ and $\dot{x} = 2t$

$1 = 2t$

$t = 0.5$ s

9 The velocity v m s^{-1} at time t seconds $(t \geq 0)$ of a body moving in a straight line is given by $v = 6t^2 + 6t - 12$. Find:

a the acceleration when the velocity is zero.

Velocity 0 at $t = 1, a = 18$ m/s²

b the initial velocity and acceleration.

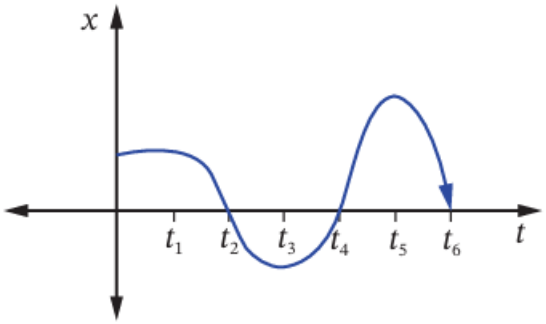
At $t = 0$, $v = -12$ m/s, $a = 6$ m/s²

10 The graph shows the displacement of a particle as it moves along a straight line.
When is the particle:

a at the origin?

b at rest?

c furthest from the origin?



11 This graph shows the velocity of a particle.

a When is the particle at rest?

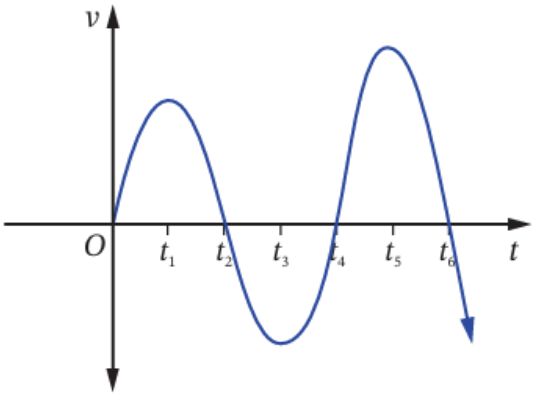
b When is the acceleration zero?

c When is the speed the greatest?

d Describe the motion of the particle at:

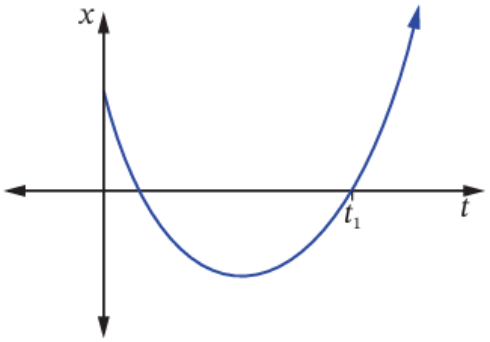
i t_2

ii t_3

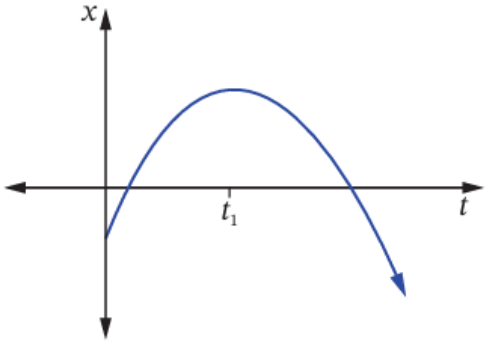


12 Describe the displacement and velocity of the particle at t_1 for each displacement graph.

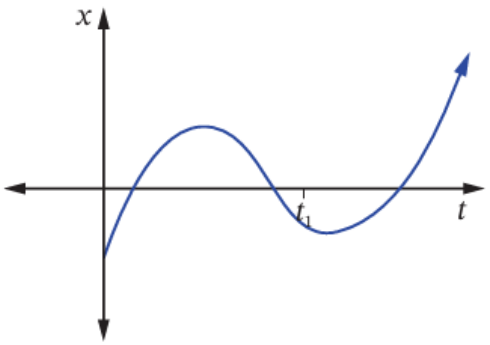
a



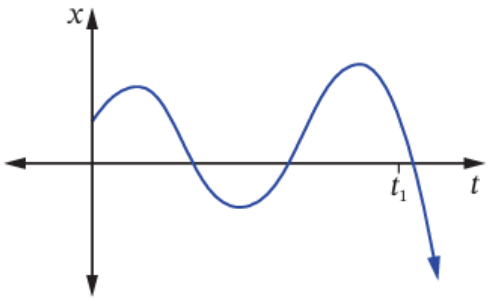
b



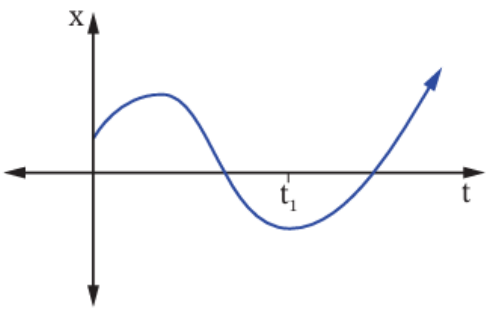
c



d

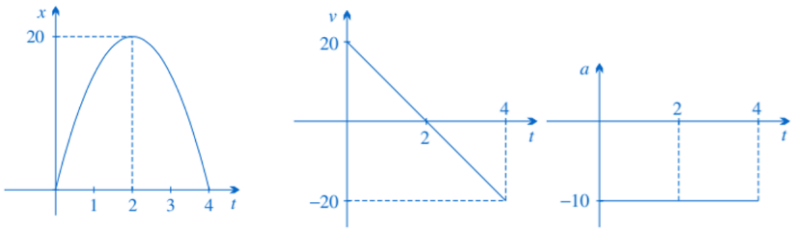


e



13 A cricket ball is thrown vertically upwards. Its height x in metres at time t seconds after it is thrown is given by $x = 20t - 5t^2$.

a Find v and a as functions of t, and show that the ball is always accelerating downwards. Then sketch graphs of x, v and a against t.



$x = 5t(4 - t)$

$v = 20 - 10t$

$a = -10$

b Find the speed at which the ball was thrown.

$v(0) = 20$ m/s

c Find when it returns to the ground (that is, when $x = 0$) and show that its speed then is equal to the initial speed.

It returns to the ground at $t = 4$. At both times, the speeds are 20 m/s

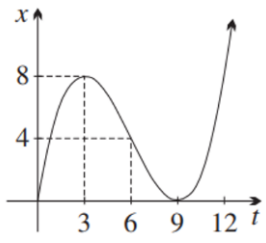
d Find its maximum height above the ground and the time taken to reach this height.

Max height at $t = 2$ s, $h = 20$ m

e Find the acceleration at the top of the flight, and explain why the acceleration can be non-zero when the ball is stationary.

-10 m/s². Although the ball has stopped, its velocity is changing, meaning its acceleration is non–zero.

14 A particle is moving horizontally so that its displacement x metres to the right of the origin at time t seconds is given by the graph.



a In the first 10 seconds, what is its maximum distance from the origin and when does it occur?

8 metres at $t = 3$ s

b By examining the gradient, find when the particle is:

i stationary,

$t = 3, t = 9$

ii moving to the right,

Moving to the right when derivative is positive.
$0 \leq t < 3$ and $t > 9$ (gradient is positive)

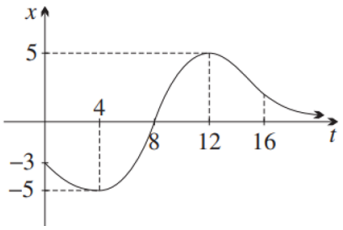
iii moving to the left.

Moving to the left when derivative is negative.
$3 < t < 9$ (gradient is negative)

c When does it return to the origin and what is its velocity then?

$t = 9$, velocity is 0.

15 A particle is moving vertically according to the graph shown, where upwards is positive.



a At what times is this particle:

i below the origin?

$0 \leq t < 8$

ii moving downwards?

$0 \leq t < 4$ and $t > 12$

b At about what time is its speed greatest?

Gradient steepest at $t \approx 8$

c At about what time is:

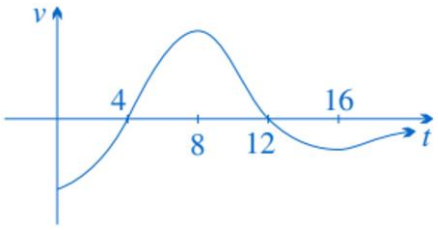
i the distance from the origin about the same as at $t = 3$?

$t \approx 5, 11, 13$

ii the velocity about the same as at $t = 3$?

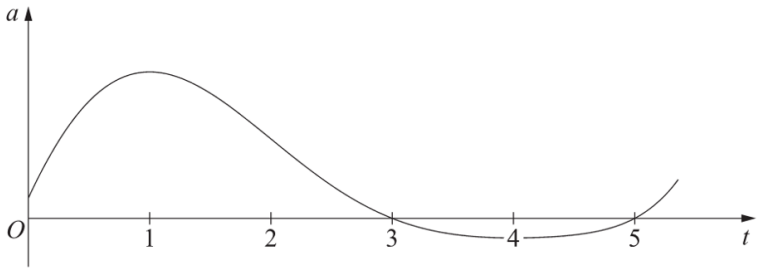
Gradient of tangent about the same when $t \approx 13$ and $t \approx 20$

d Draw an approximate sketch of the graph of v as a function of time.



16 **2019 HSC Advanced Band 4**

A particle is moving along a straight line. The graph shows the **acceleration** of the particle.



For what value of t is the **velocity** v a maximum?

A. 1

B. 2

C. 3

D. 5

C

Velocity is at min or max when $a = 0$

At $t = 3$, acceleration changes from positive to negative, so velocity changes from increasing to decreasing.

Therefore, maximum velocity at $t = 3$.

In contrast, $t = 5$ would be minimum velocity.